

Ekam Bhullar

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EDUCATION

Purdue University

May 2028

B.S. Computer Engineering

West Lafayette, IN

Relevant coursework: data structures & algorithms (C), digital design (SystemVerilog), signals & systems, microprocessor systems.

EXPERIENCE

Undergraduate Researcher, Computational Quantum Electromagnetics Lab

Aug–Dec 2025

Under Dr. Roth, Purdue University

- Engineered a Python automation framework for GMSH to streamline 3D mesh generation, reducing manual Abaqus input file preparation time by 50–70%.
- Developed modular scripts for conformal meshing and physical group assignments, producing high-fidelity FEM-ready structures.
- Authored technical documentation ensuring 100% reproducibility for future large-scale simulations.

Undergraduate Researcher & OUR Scholar, re3data Recommendation Engine

Aug 2025–May 2027

Under Prof. Witt, Purdue University

- Selected as an OUR Scholar with a \$1,000 research scholarship for AI-enhanced metadata curation of the re3data repository.
- Architected a high-concurrency Flask backend serving content-based recommendations for 3,400+ research repositories, reducing search discovery time by 30%.
- Built semantic search using TF-IDF and cosine similarity, validated via t-SNE clustering.
- Implemented a reliability-aware API layer with real-time HTTP status checks, increasing accessible repository visibility by 40%.

Teaching Assistant, Engineering in the World of Data

Aug 2025–May 2026

Purdue University

- Guided first-year students through the full data lifecycle; managed grading and delivered structured feedback.

PROJECTS

Quantum Simulation Engine

Jun 2026–Present

C++17/20 · numerical methods

- Built a from-scratch C++ simulation engine — ODE/PDE solvers (RK4, Crank-Nicolson) and a quantum circuit simulator validated via Bell-state and teleportation protocols (1000/1000 trials)
- Engineered a CUDA state-residency optimization, eliminating per-gate transfer overhead for a 95.9× speedup at 256 gates, verified bit-exact (0.000e+00 error) against CPU
- Applied the roofline model across OpenMP and CUDA kernels — 3.2× compute-bound speedup, diagnosed a transfer-bound GPU regime (96% PCIe overhead) that drove the residency redesign

esh — Unix Shell

Summer 2026

C · POSIX

- Fully functional Unix shell using only POSIX system calls: fork/exec/wait, pipes, dup2 I/O redirection, and signal handling for Ctrl+C.
- No external frameworks; demonstrates direct understanding of OS internals and the hardware-software boundary.

Multi-Client TCP Chat Server

Summer 2026

C · socket programming

- Real-time TCP server using select-based I/O multiplexing for concurrent clients in a single-threaded, event-driven architecture (the same model as nginx and Redis).
- Stateful sessions, message broadcasting, and private messaging via command parsing, with a custom TCP client.

Exoplanet Database & Analysis Engine

Summer 2026

PostgreSQL · Node.js/Express · Chart.js

- Full-stack app analyzing 5,700+ NASA-confirmed exoplanets; PostgreSQL with window functions, CTEs, and aggregations verifying Kepler's third law and computing stellar habitable zones.
- REST API with multiple analytics endpoints; frontend visualizations with Chart.js.

SKILLS

Languages: C/C++17/20, Python, SQL, JavaScript (Node.js), SystemVerilog, MATLAB, Bash

Systems: POSIX APIs, socket programming, I/O multiplexing, process management, finite-difference methods, OpenMP, CUDA

Tools: PostgreSQL, Express, Flask, NumPy, Pandas, scikit-learn, Chart.js, Git, Linux/WSL, Valgrind/GDB

Concepts: OS internals, numerical methods, ODE/PDE solvers, GPU memory optimization, FEM mesh generation, statistical modeling

WRITING

The wave equation is a time machine

Medium · Jun 2026

Original derivation and physical interpretation of D'Alembert's wave equation discretization.

COMPETITIONS & COURSES

- IMC Prosperity 4 — algorithmic trading competition, 2026
- Akuna Capital Options 101 — options theory and derivatives coursework, completed 2026